Pharia — The Galaxy Screen: Master UI & Interaction Spec

> **Scope:** The phone-based spatial discovery screen ("the Galaxy") only. This is the single most important screen in the product and the centerpiece of the demo.
> **How to use this doc:** Part 1 (Visual) can be pasted into an AI mockup tool (Uizard/Visily) or read by a designer — it describes static appearance. Part 2 (Interaction) is for developers / Claude Code — it describes every gesture, state, and transition that a static mockup can't capture. Part 3 is the data the screen needs.

PART 1 — VISUAL DESCRIPTION (the static look)

1.1 Canvas & frame

- Target device: phone, portrait. Design at **393 × 852** (iPhone 15/16) or equivalent Android.
- The screen is **edge-to-edge dark**. Background base color `#07070c` (near-black, very slightly blue-violet).
- Behind everything, a **soft radial glow** sits roughly centered, slightly below middle (around 50% horizontal, 55% vertical). The glow color matches the current mood region — by default violet `#b56cff` at ~18% opacity at its core, fading to transparent by ~60% radius. This glow is the single most important atmospheric element; it makes the screen feel alive rather than flat.
- Optional: a barely-perceptible film grain overlay (~3–4% opacity) over everything for texture.

1.2 Three fixed layers, front to back

The screen has three stacked layers. From back to front:

1. **The galaxy world** (back) — the pannable space containing show cards, connection lines, and ambient dots. This layer MOVES when the user explores.
2. **The fixed overlay** (middle) — compass labels at the four edges, the user-position indicator at center. These DO NOT move when panning.
3. **The chrome** (front) — status bar, mood pill (top), queue bar (bottom). These are app UI, always fixed, always on top.

1.3 Top chrome

- **Status bar** (very top, ~16px down, left and right insets ~22px): left shows time "9:41" in JetBrains Mono ~10px, ink-soft color `#c8bfa8`. Right shows signal/network glyphs, same treatment. This is cosmetic (mimics OS status bar).
- **Mood pill** (centered horizontally, ~52–60px from top): a pill-shaped chip. Background `rgba(255,255,255,0.04)`, 1px border `rgba(255,255,255,0.10)`, fully rounded (100px

radius), padding ~6px 12px. Inside: a small pulsing dot (5px, colored to current mood — violet by default) followed by mono-uppercase text "COZY · REFLECTIVE" at ~8px, letter-spacing 0.16em, ink color. The dot gently pulses (opacity + scale) on a ~2s loop.

1.4 The galaxy world (the content)

This is the heart of the screen. It is a **larger-than-viewport space** (think ~800 × 900 world inside a ~393-wide window) that the user pans through. At rest, the viewport is centered on the world center.

Mood regions — the world is organized into five emotional zones by direction:

- **Center: Cozy / Reflective** — accent violet `#b56cff`
- **North (up): Cerebral** — accent blue `#6fa8e8`
- **East (right): Nostalgic** — accent amber `#e6b04a`
- **South (down): Escape** — accent teal `#3fb89e`
- **West (left): Intense** — accent red `#ff7a6c`

Show cards — each show is a small **poster card**, a rounded rectangle (radius ~5px, 1px border `rgba(255,255,255,0.18)`), 2:3 aspect ratio (portrait, like a movie poster). Cards carry a soft drop shadow plus a faint glow tinted to their mood region.

- Card sizes vary to imply depth and relevance: focal/large (~60–64px wide), medium (~46–48px), small (~36–38px), extra-small/dim (~28–30px, reduced opacity).
- Cards near the center (the current mood) are **larger and brighter**; cards toward the edges (other moods) are **smaller and dimmer**. This communicates "you are here; those are far away."
- Each card's artwork is an **abstract gradient / simple shape** evoking the show's tone (NOT a real poster — copyright-safe, on-brand). E.g. two figure silhouettes for a romance, geometric squares for a cerebral sci-fi, warm sunset gradient for a reflective drama.
- Below each card, a tiny **label** in JetBrains Mono ~6–7px, ink at ~70% opacity, the show title in caps (e.g. "PAST LIVES"). Medium/large cards always show labels; smallest cards may only show on hover.

Connection lines — faint hairlines (0.5px, mood-tinted, ~8–30% opacity, animated to subtly pulse) connect a few cards in the central cluster, implying "these are similar." Decorative, low-key.

Ambient dots — scattered tiny white/colored dots (0.5–1px) at low opacity, twinkling slowly, to give the sense of stars/depth in the empty space.

Example content layout (world coordinates, illustrative):

- Center cluster (Cozy): Past Lives (large, focal), Normal People, Fleabag, Aftersun, Lady Bird
- North (Cerebral): Severance, Westworld, Dark, Mr. Robot, Devs
- East (Nostalgic): Stranger Things, Mad Men, That '70s Show, Bridgerton, Freaks and Geeks

- South (Escape): Schitt's Creek, Ted Lasso, The White Lotus, Emily in Paris, Loot
- West (Intense): Succession, Breaking Bad, The Wire, The Bear, Better Call Saul

1.5 The fixed overlay

- **Compass labels** at the four screen edges, each tinted to its mood and slightly translucent (~55% opacity), brightening on hover/proximity:
  - Top center: "↑ CEREBRAL" (blue), arrow above text, stacked vertically
  - Right center: "NOSTALGIC →" (amber), horizontal
  - Bottom center: "ESCAPE ↓" (teal), text above arrow, stacked
  - Left center: "← INTENSE" (red), horizontal
  - All in JetBrains Mono ~8px, letter-spacing 0.22em, uppercase, with a subtle dark text-shadow so they stay legible over cards.
- **User-position indicator** at the exact center of the viewport: a **dashed circle** (~88px diameter, 1px dashed `rgba(255,255,255,0.18)`) slowly rotating (~28s per revolution), with a second fainter solid ring inset inside it. A tiny "YOU" label in mono ~6px sits just below it. This never moves; the world moves behind it.

1.6 Bottom chrome — the queue bar

- A pill/rounded-rectangle bar pinned near the bottom (~18px inset left/right/bottom), radius ~12–16px. Background `rgba(8,7,13,0.8)` with backdrop blur, 1px border `rgba(255,255,255,0.10)`.
- Left side: a tiny mono label "TONIGHT'S QUEUE" (~7px, ink-dim, uppercase), and below it a count in Instrument Serif italic ~16px, ink: "3 picks ready".
- Right side: a mono "SEND →" affordance (~8–9px, letter-spacing 0.16em, ink-soft) — taps to push the queue to the TV.

1.7 Typography & color summary

- **Display / titles:** Instrument Serif (use italic for emphasis). Warm, editorial.
- **Body:** Geist (or Inter fallback).
- **Labels / mono / eyebrows:** JetBrains Mono, uppercase, wide letter-spacing.
- **Ink:** `#f4ebd9` (warm off-white). Ink-soft `#c8bfa8`. Ink-dim `#6a6457`.
- **Moods:** Cozy `#b56cff` · Cerebral `#6fa8e8` · Nostalgic `#e6b04a` · Escape `#3fb89e` · Intense `#ff7a6c`.
- Overall feel: **atmospheric, cosmic, A24-film-poster sophistication. NOT generic SaaS.** Generous negative space, glassmorphism on overlays, ambient glows.

PART 2 — INTERACTION MODEL (every gesture, state, transition)

This is the part a static mockup cannot express. Each interaction below specifies the trigger, the response, the animation, and the resulting state.

2.1 The core mental model

The user is an explorer at a fixed point (viewport center). The **world moves around them**, not the other way around (direct-manipulation, like a map). Panning reveals different mood regions. The current mood is whatever region is nearest the viewport center. Everything else keys off this.

2.2 State variables the screen tracks

- `viewX, viewY` — current look-position offset from world center (px). `(0,0)` = looking at Cozy center.
- `targetX, targetY` — where an animated pan is heading (for momentum/compass jumps).
- `velX, velY` — drag velocity, for momentum on release.
- `isDragging` — boolean.
- `currentMood` — derived from `viewX/viewY` (see 2.6).
- `queue[]` — array of queued show IDs (starts with a seeded count, e.g. 3).
- `hoveredCard` / `activeDetailCard` — which card, if any, is hovered or showing its detail popup.
- `panBounds` — clamp limits, e.g. `PAN_MAX_X ≈ 260`, `PAN_MAX_Y ≈ 270`, so the user can reach every region but not pan into empty void.

2.3 Drag to pan (primary navigation)

- **Trigger:** pointer/touch down anywhere on the galaxy that isn't a card, then move.
- **On down:** set `isDragging = true`, record start pointer position and start `viewX/viewY`, add a `.dragging` class (cursor → `grabbing`; disables the world's smooth-transition so it tracks the finger 1:1), hide any open detail popup, hide the first-run pan hint.
- **On move:** `viewX = startViewX - (pointerX - startX)`; `viewY = startViewY - (pointerY - startY)` (direct manipulation — drag right reveals the western/left world). Clamp both to `panBounds`. Continuously track velocity from the last few move deltas.
- **On up/release:** set `isDragging = false`, remove `.dragging`. Apply **momentum**: set `targetX/targetY = viewX/viewY + velocity × decayFactor`, clamped to bounds. The world glides to target with an ease-out curve (~0.8–1.2s) then settles.
- **Touch:** same logic, single-finger. Use passive listeners where possible. Ignore multi-touch (no pinch in v1).

2.4 Compass tap (jump-to-region)

- **Trigger:** tap a compass label (e.g. "↑ CEREBRAL").

- **Response:** animate `targetX/targetY` to that region's center (e.g. North → `(0, -PAN_MAX_Y)`), world glides over ~0.9–1.2s with `cubic-bezier(0.2,0.7,0.2,1)`.
- During and after the glide, `currentMood` updates live, which drives the mood pill text + dot color + glow color (see 2.6).
- **Toggle behavior:** if the user is already centered on that region and taps the same compass, return home to center `(0,0)`. (Compass acts as a toggle: out and back.)
- The tapped compass label brightens to full opacity while active.

2.5 Cursor / device parallax (depth feel) — optional but desired

- **Trigger:** pointer moves over the galaxy without dragging (desktop), or device tilt (mobile, if you wire gyroscope — optional).
- **Response:** cards shift by a tiny amount opposite the cursor, scaled by a per-card depth factor (closer/larger cards move more, distant/smaller cards move less), giving a subtle 3D parallax. Magnitude small (a few px). Eased, never jittery. Disabled while dragging.

2.6 Mood derivation & ambient response (the signature behavior)

- Continuously compute `currentMood` from `viewX/viewY`:
  - If distance from center `< ~110px` → **Cozy**.
  - Else by angle: right quadrant → Nostalgic, down → Escape, left → Intense, up → Cerebral.
- When `currentMood` changes, animate three things together over ~0.6s:
  1. **Mood pill text** swaps (e.g. "CEREBRAL · ANALYTICAL").
  2. **Mood pill dot color** crossfades to the new mood accent.
  3. **Background radial glow color** crossfades to the new mood accent (this is the big atmospheric payoff — the whole screen's ambient color shifts as you explore).
- (Stretch goal) **Ambient audio**: a low pad/drone crossfades between mood-specific tones. Optional for demo; powerful if it works. Mute by default with a toggle.

2.7 Card hover (desktop) / dwell

- **Trigger:** pointer enters a card and stays (not dragging).
- **Immediate response (on enter):** card scales up (~1.18×) and lifts slightly (translateY -3px), brightness up, z-index raised above neighbors. Its label appears if it was hidden. Eased ~0.4s.
- **After a delay (~450ms of continuous hover):** the **detail popup** opens (see 2.8).
- **On leave:** card returns to rest; pending popup timer cancels; open popup hides.

2.8 The detail popup (long-hover / tap-and-hold)

- **Trigger:** ~450ms hover on desktop, OR tap-and-hold on touch.

- **Appearance:** a small glassmorphism panel (~168px wide), background `rgba(14,12,24,0.92)` + backdrop blur, 1px border, soft shadow + mood glow. Contains:
  - Show title (Instrument Serif ~16px)
  - Meta line (mono ~7px): year · runtime · genres
  - One-sentence description (~10px)
  - Footer row: match score ("96%" with serif numeral + "match" label) on the left; a mood badge pill (colored to the show's mood) on the right
  - A "+ TAP TO QUEUE" affordance
- **Smart positioning:** prefer to the right of the card; if it'd overflow the right edge, place left; if that overflows, center horizontally and clamp vertically — always fully on-screen, ~8px from edges.
- **Animation:** scale from 0.92→1 + fade in over ~0.25–0.3s, transform-origin toward the card.
- **Dismiss:** instantly on pointer leave / drag start / queue action / tapping elsewhere.

2.9 Queue a show (the core conversion action)

- **Trigger:** tap a card (quick tap, not a drag — distinguish by movement threshold, e.g. <5px since pointer-down = tap), OR tap "+ TAP TO QUEUE" in the popup.
- **Guard:** if already queued, do nothing (or allow un-queue — designer's call; v1 = no-op).
- **Response (the satisfying micro-moment):**
  1. Mark card `queued`: its border switches to the violet accent, glow intensifies, and a small ✓ badge appears at the top-right corner of the card.
  2. A "+ QUEUED" chip **flies** from the card toward the queue bar at the bottom (animate translate down + fade out over ~1.4s, ease-out).
  3. The **queue count increments** ("3 picks ready" → "4 picks ready"), with the serif numeral doing a quick pop.
  4. The **queue bar pulses** (brief glow/box-shadow flash ~0.6s) to confirm receipt.
- Dismiss any open detail popup.

2.10 Send to TV (the handoff trigger)

- **Trigger:** tap "SEND →" on the queue bar.
- **Response (demo):** a confirmation state — queue bar briefly shows "Sent to your TV ✓", or a full-screen transitional beat. In the real product this writes the queue to the cloud; for the demo it can trigger the mocked TV screen / screen-recording. This is the bridge to the magic moment.

2.11 First-run / empty states

- **Pan hint:** on first load, a faint pulsing "DRAG TO EXPLORE" hint near the lower-center, mono ~7px. It fades out permanently the moment the user drags or taps a compass.

- **Cold start (no taste data):** the galaxy still renders with a default/seeded layout (popular + curated picks) so it's never empty. An onboarding mood-tap (out of scope here) seeds the starting region.
- **Loading:** if data is async, show the ambient glow + a few skeleton card shimmers in the center cluster, then populate.

2.12 Idle behavior

- When untouched, cards **drift gently** (each on a slightly different slow loop, ~5–8s, a few px of vertical bob), the user ring rotates, ambient dots twinkle, connection lines pulse. The screen is never fully still — it breathes. All idle motion pauses or yields during active drag.

2.13 Transitions in/out of the screen (Combo 01 context)

- The Galaxy lives **one swipe below the Daily Card** (the front door). Swiping up from the Daily Card reveals the Galaxy with a smart-animate/parallax transition. Swiping down from the top of the Galaxy returns to the Daily Card. (Daily Card itself is specced separately; noting the seam here.)

2.14 Accessibility & reduced motion

- Respect `prefers-reduced-motion`: disable idle drift, parallax, and the rotating ring; keep functional transitions near-instant. Discovery must still work without motion.
- Compass labels and queue actions must be reachable/tappable with adequate hit targets (⩾44px effective).
- Maintain text contrast over the dark field; the text-shadows on labels exist for this reason.

PART 3 — DATA THE SCREEN NEEDS

Minimal shape per show (for mockups, hardcode ~20–25 of these):

```
Show {
  id: string
  title: string            // "Past Lives"
  year: number             // 2023
  runtime: string          // "1h 46m" or "2 seasons"
  genres: string           // "Drama · Romance"
  description: string      // one sentence
  match: number            // 0–100, e.g. 96
  mood: "cozy" | "cerebral" | "nostalgic" | "escape" | "intense"
  worldX: number           // position in galaxy space
  worldY: number
  size: "l" | "m" | "s" | "xs"   // relevance/depth tier
  artwork: <abstract gradient/shape spec or asset ref>   // NOT a real poster
}
```

Screen-level state:

```
GalaxyState {
  viewX, viewY: number
  currentMood: MoodKey
  queue: string[]          // show IDs
  hoveredCardId: string | null
  detailCardId: string | null
}
```

For the demo, the **recommendation logic** (which shows appear, where, and their match scores) can be precomputed/hardcoded, or generated by feeding a fake viewing-history JSON + a mood to an LLM and placing the returned shows. Either way, the screen just renders whatever `Show[]` it's handed.

Build notes

- The world-moves-not-the-user model is the key implementation decision; get the pan math right first, everything else layers on top.
- The mood-glow crossfade (2.6) is the single highest-impact piece of polish. Prioritize it.
- The queue micro-moment (2.9) is the core conversion feedback. Make it feel good.
- Idle breathing (2.12) is what separates "looks like a mockup" from "feels alive" — cheap to add, big payoff.

End of spec.